

★ WEEK 9 · PART 2 of 3

The Kalman Gain and the Full Recursion

The Number That Balances Model Against Data

The gain formula · the covariance P · the five equations · optimality

The update step corrects by the gain times the surprise. Part 2 derives that gain, shows how it tracks the filter's own uncertainty, assembles the full recursion, and states exactly when it is optimal.

2

PART 2 • THE GAIN

The Heart of the Filter

The Kalman gain and what it balances, the covariance that drives it, the full five-equation recursion assembled in one place, and the conditions under which the filter is optimal.

We Have the Cycle — Now, the Gain

Part 1 gave the two steps; Part 2 gives the weight

1

The correction is gain times surprise

Part 1 showed the update corrects the state by K times the innovation. But we left the gain K as a symbol — the one piece still missing.

2

The gain is the whole balancing act

It decides how much to trust the new data versus the model's prediction. Get it right and the estimate is optimal; get it wrong and you over- or under-react.

3

It falls out of the uncertainties

Remarkably, the optimal gain is determined entirely by the covariances the filter already tracks. No tuning — the math hands it to you.

Part 2 in one line: the gain optimally weighs model against data — and the filter computes it for free.

The Gain: The Heart of the Filter

How much to trust the new observation

Every correction the filter makes is scaled by the Kalman gain. It is a single number (or matrix) that answers, at every step: given what I know and how noisy the data is, how far should I move toward this observation?

What it controls

- The size of the correction in the update step.
- The balance between the model and the new data.
- How quickly the estimate reacts to change.

Why it is the heart

- Everything else is bookkeeping; the gain is the decision.
- It is where the filter's intelligence lives.
- The optimal gain is what makes the Kalman filter Kalman.

The Kalman Gain

State uncertainty over total uncertainty

$$K_t = P_{t|t-1} Z' F_t^{-1}, \quad F_t = Z P_{t|t-1} Z' + H$$

Reading it

- The numerator is the predicted state uncertainty.
- F , the denominator, is the innovation variance.
- F combines state uncertainty and measurement noise H .

The message

- The gain is a RATIO of uncertainties.
- More state uncertainty $\rightarrow$ larger gain $\rightarrow$ trust the data.
- More measurement noise $\rightarrow$ smaller gain $\rightarrow$ trust the model.

The Gain in One Dimension

A number between zero and one

$$\text{scalar case: } K = \frac{P_{\text{pred}}}{P_{\text{pred}} + H} \in [0, 1] \quad (\text{state unc.})/(\text{total unc.})$$

What it means

- In the scalar case, the gain lies between 0 and 1.
- It is the fraction of the surprise you keep.
- A gain of 0.4 moves the estimate 40% of the way to the data.

The extremes

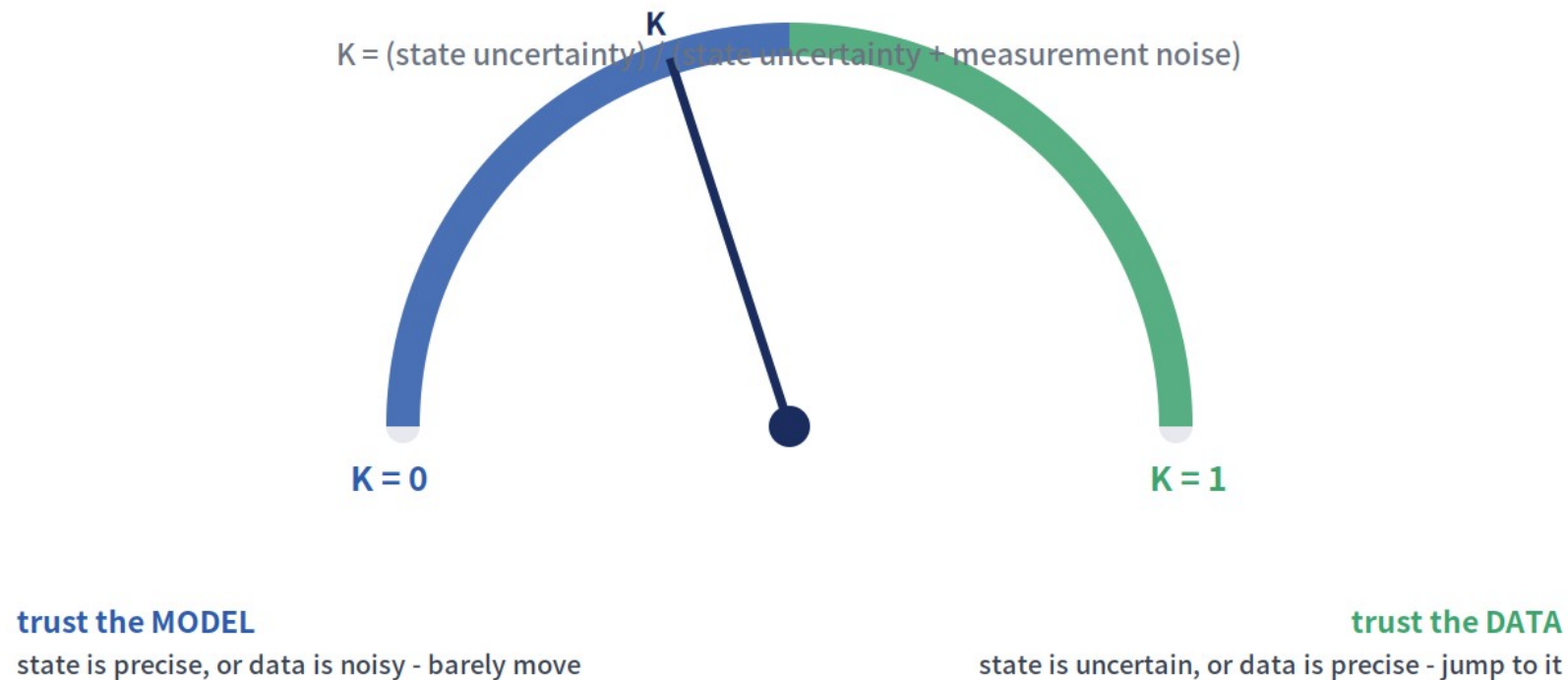
- Near 1: the data dominates — you nearly adopt the observation.
- Near 0: the model dominates — you barely budge.
- The filter picks the point that minimizes error.

The Gain as a Trust Dial

The gain is a dial from 0 to 1: trust the model when the state is precise or the data is noisy; trust the data when the state is uncertain or the measurement is sharp.

The Gain as a Trust Dial

The Kalman gain K is a number between 0 and 1 - how far to move toward the new data.



Trust the Model or Trust the Data?

The gain answers this at every step

The gain automatically resolves the central tension of estimation. When the model is more reliable than the data, it leans on the model; when the data is sharper, it leans on the data. No manual tuning.

When the model wins

- The state estimate is already precise (small P).
- Or the measurement is very noisy (large H).
- The gain is small — the observation barely moves you.

When the data wins

- The state estimate is uncertain (large P).
- Or the measurement is precise (small H).
- The gain is large — you jump toward the observation.

The Extremes of the Gain

Two behaviors at the ends of the dial

$K \rightarrow 1$: trust the DATA

$K \rightarrow 0$: trust the MODEL

Gain near one

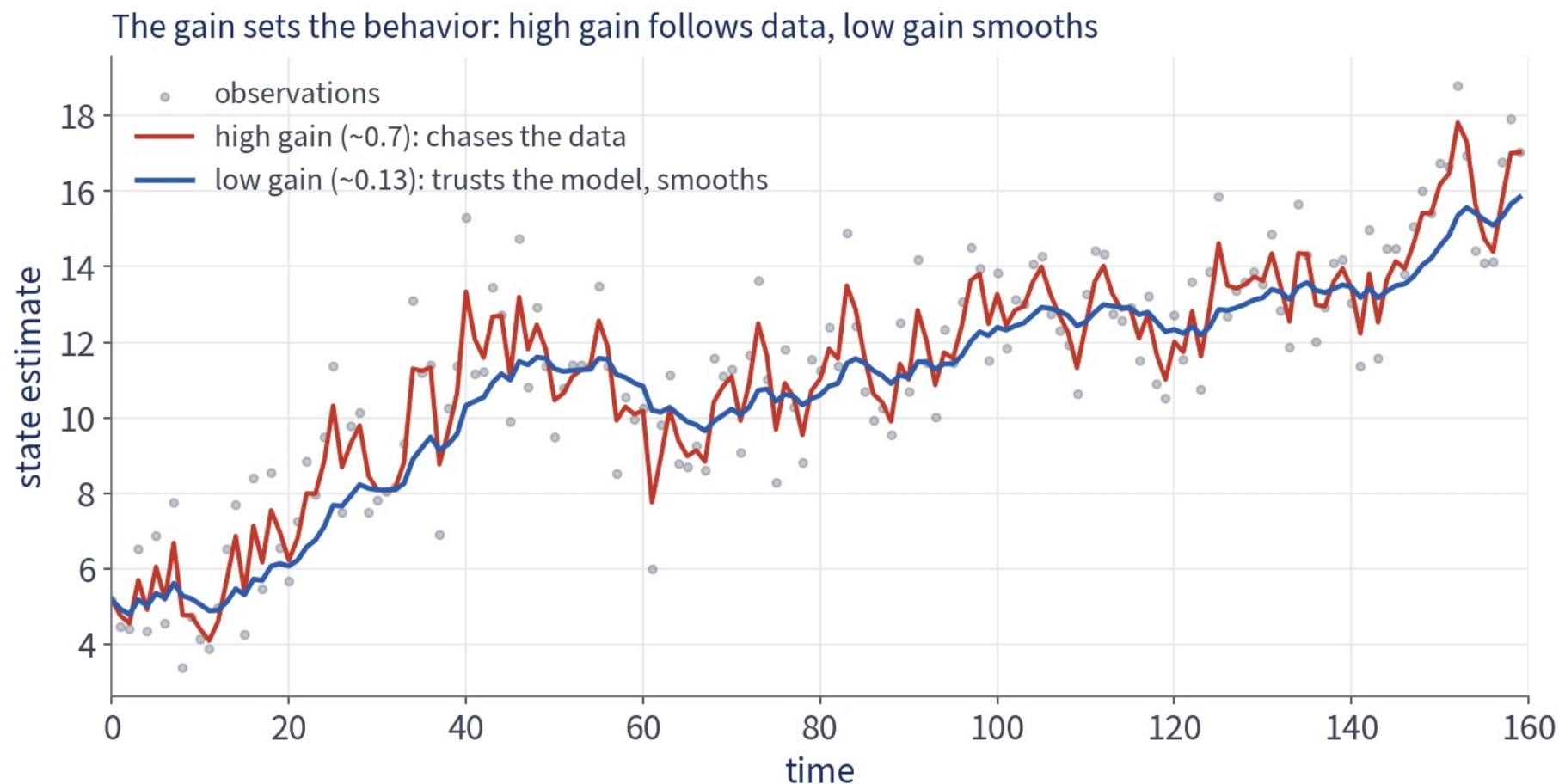
- The filter essentially adopts each new observation.
- It tracks fast but inherits the data's noise.
- Useful when the state truly moves quickly.

Gain near zero

- The filter mostly ignores the data and trusts the model.
- It is very smooth but slow to react.
- Useful when the state is nearly constant.

The Gain, in Action

Two filters on the same data: a high gain chases every observation (noisy but responsive); a low gain trusts the model and smooths (stable but slower). The gain sets the whole character.



The Covariance P

The filter knows how sure it is

covariance update: $P_{t|t} = (I - K_t Z) P_{t|t-1}$ (always shrinks)

What P is

- The covariance of the state estimate — its uncertainty.
- It grows in predict (by Q), shrinks in update.
- The update always reduces it: data adds information.

Why it matters

- P feeds directly into the gain each step.
- It gives calibrated error bars on every estimate.
- The filter literally tracks its own confidence.

Why the Update Shrinks P

Information can only help

The logic

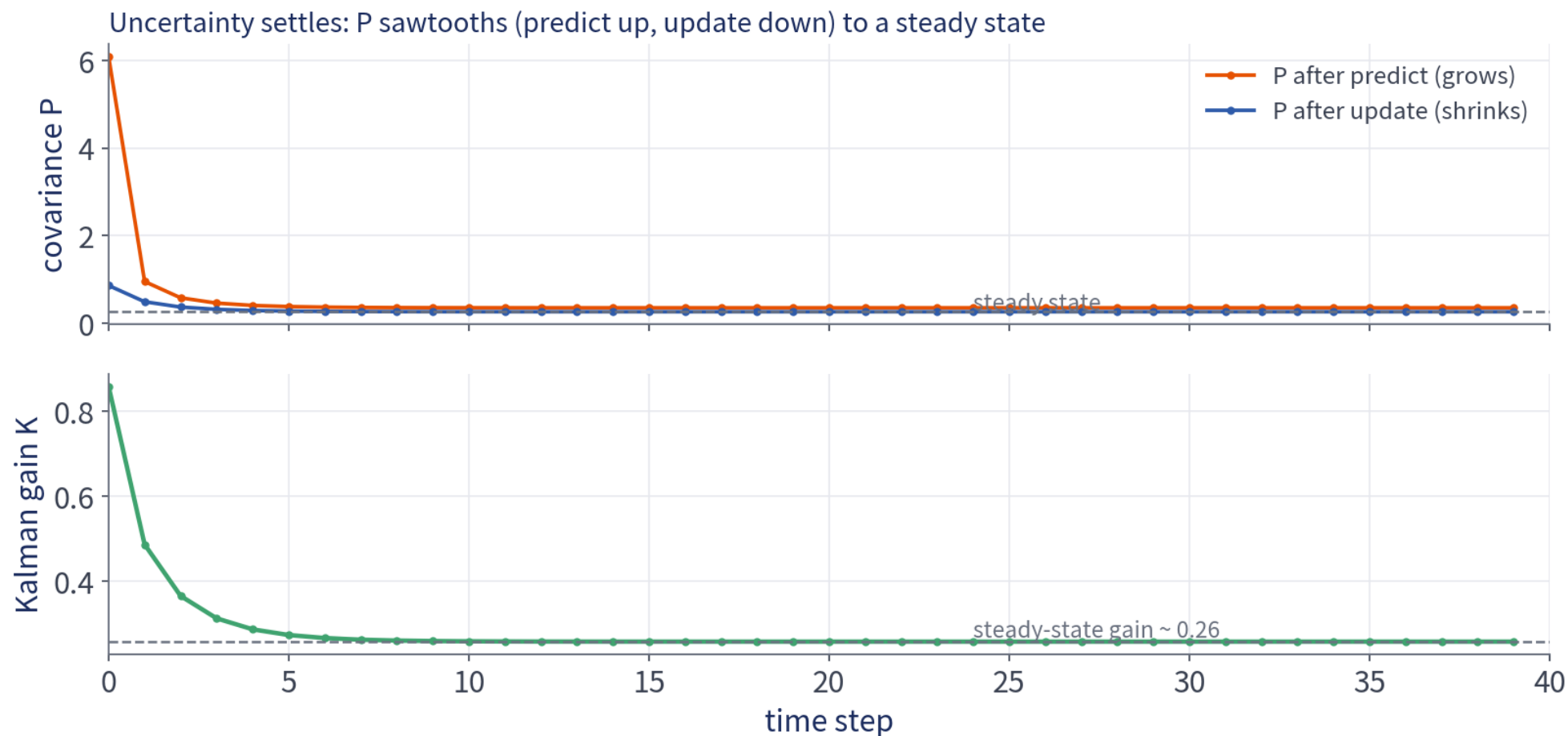
- An observation adds information about the state.
- More information means less uncertainty.
- So the update can only reduce P , never increase it.

The balance with predict

- Predict adds process noise Q — uncertainty grows.
- Update removes uncertainty — it shrinks.
- The two forces reach a balance over time.

Uncertainty and Gain Settle

The covariance sawtooths — up in predict, down in update — and converges (top). The gain converges with it to a steady-state value (bottom). A time-invariant filter reaches equilibrium.



The Gain Adapts as P Changes

High early, then settling

Early on

- The initial state is uncertain, so P is large.
- That makes the gain high — the filter trusts data heavily.
- It converges quickly toward the truth.

Later

- As information accumulates, P shrinks and stabilizes.
- The gain settles to a steady value.
- The filter reaches a stable trust between model and data.

The Full Recursion at a Glance

Five equations, two steps — assembled

$$\begin{array}{ll} \text{PREDICT} & \hat{x}_{t|t-1} = T \hat{x}_{t-1|t-1} \quad P_{t|t-1} = T P_{t-1|t-1} T' + Q \\ \text{innovation} & v_t = y_t - Z \hat{x}_{t|t-1} \quad F_t = Z P_{t|t-1} Z' + H \\ \text{gain} & \\ \text{UPDATE} & \hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t v_t \quad P_{t|t} = (I - K_t Z) P_{t|t-1} \end{array}$$

The two steps

- Predict: move the state and grow P.
- Update: gain, then correct the state and shrink P.
- The innovation v and its variance F link the two.

How to use it

- Memorize the shape, not the matrix algebra.
- Predict has no data; update is where data enters via K .
- Everything reduces to these five lines.

The Five Equations, Named

Each line, and its job

Step	Equation	What it does
Predict state	$x = T x$	move the state forward
Predict cov	$P = T P T' + Q$	grow the uncertainty
Gain	$K = P Z' / F$	weigh model vs data
Update state	$x = x + K v$	correct by gain times surprise
Update cov	$P = (I - K Z) P$	shrink the uncertainty

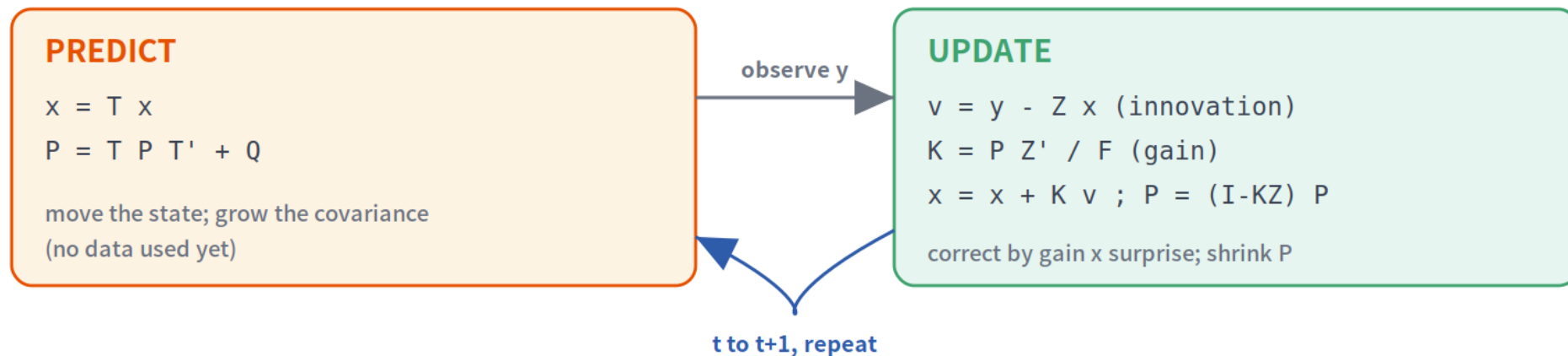
$F = Z P Z' + H$ is the innovation variance; $v = y - Z x$ is the surprise. That is the entire filter.

The Full Recursion, in the Loop

The five equations placed in the predict-update loop. Two carried objects — the state x and covariance P — summarize the entire past; the gain K is the only place data enters.

The Full Recursion, in the Loop

Five equations, two steps - each observation drives one full pass.



Two objects carry the whole past

The state estimate x and its covariance P are all you carry - a complete, compact summary. Predict inflates P by Q ; update deflates it via the gain. The gain K is the only place data enters. $F = Z P Z' + H$ is the innovation variance - the denominator that scales the gain.

When Is the Kalman Filter Optimal?

Powerful, but know the fine print

linear + Gaussian $\Rightarrow$ Kalman filter = optimal (minimum-variance) estimator

The conditions

- The system is linear (the two equations are linear).
- The noises are Gaussian.
- The matrices T , Z , Q , H are known.

What 'optimal' means

- It is the minimum-variance estimator of the state.
- It equals the exact posterior mean — nothing does better.
- Under these conditions, the Kalman filter is unbeatable.

Optimality: The Fine Print

When the assumptions bend

When they fail

- Nonlinear dynamics — the plain filter no longer applies.
- Non-Gaussian, heavy-tailed noise — optimality is lost.
- Unknown parameters must be estimated first.

The fixes

- Extended or unscented Kalman filters for nonlinearity.
- Robust or particle filters for non-Gaussian noise.
- Estimate T , Z , Q , H by maximum likelihood, then filter.

The Steady-State Gain

When the gain stops changing

The idea

- For a time-invariant system, P and K converge.
- After convergence, the gain is a fixed constant.
- You can precompute it and skip updating P each step.

Why it is useful

- It is cheaper — no covariance recursion at run time.
- It reveals the filter's long-run behavior at a glance.
- Many simple exponential smoothers are steady-state filters.

Filtering vs Smoothing, Revisited

The same filter, two ways to use it

The Kalman filter is a forward pass — it gives filtered (real-time) estimates. Add a backward pass over the same quantities and you get the smoother: the best estimate of each past state using all data.

Filtering

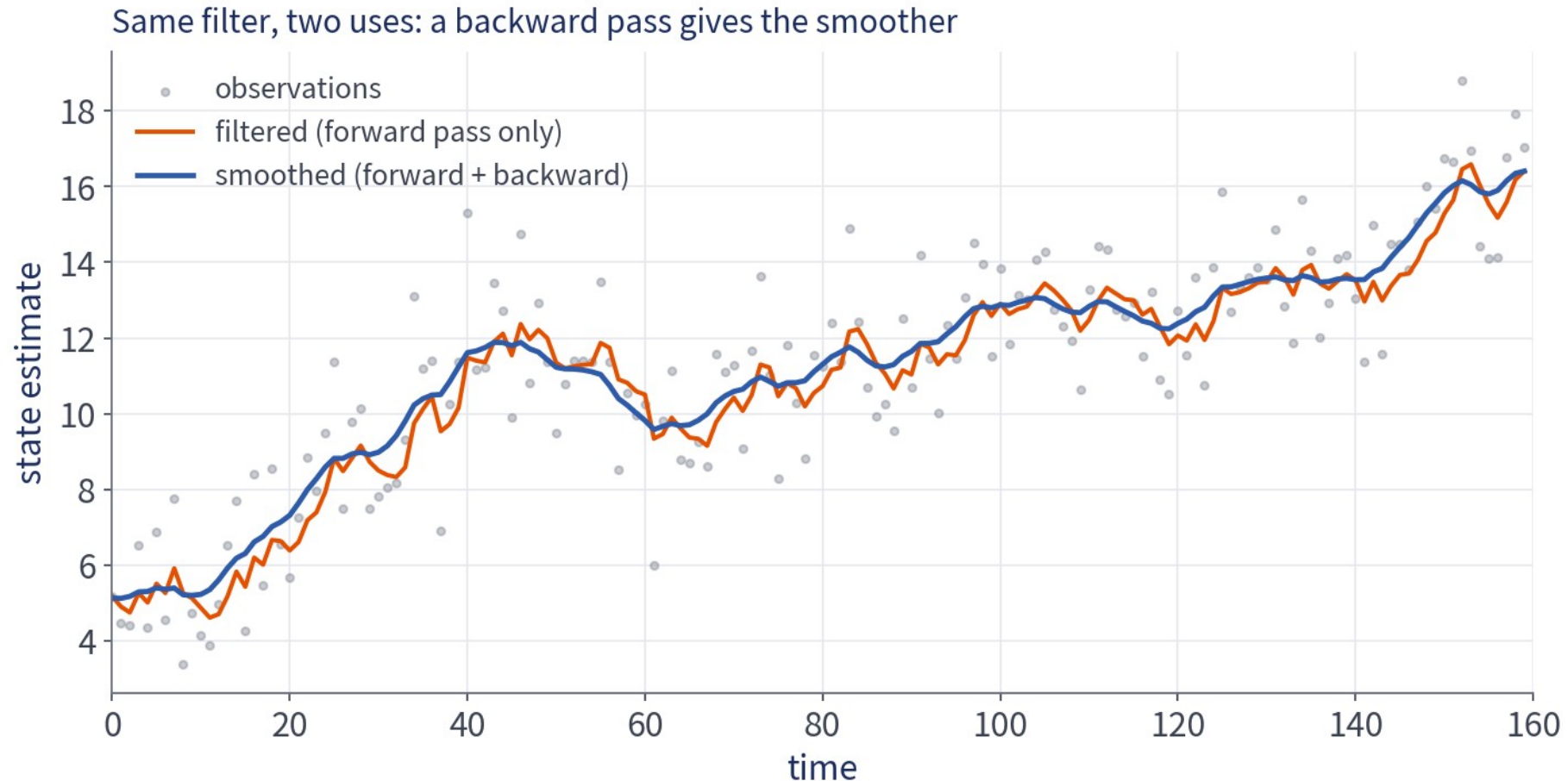
- One forward pass, causal, real-time.
- Uses data up to each point only.
- The output you trade on.

Smoothing

- A backward pass added on top of the filter.
- Uses the whole sample for every past point.
- Cleaner, but only available after the fact.

One Filter, Two Uses

The forward pass gives the filtered estimate (real-time). A backward pass over the same stored quantities produces the smoother — cleaner, using all data, but only in hindsight.



Watch the Gain and Covariance

Direct Claude Code to expose the filter's internals

GOAL Instrument a Kalman filter to see the gain and covariance evolve, and confirm they converge.

[INPUT] Prompt · Claude Code

Write and run `gain_covariance_lab.py` that:

1. runs a local level Kalman filter and records the covariance P and the gain K at every step;
 2. plots P (predicted and updated) and K over time, showing them converge to steady state;
 3. re-runs with a LARGE and a SMALL process noise Q , and compares the steady-state gains;
 4. verifies the update step always reduces P .
- Explain how a bigger Q leads to a bigger steady-state gain (trusting the data more).

How to work with it

- P should sawtooth and settle.
- K should converge to a constant.
- Bigger $Q \rightarrow$ bigger gain $\rightarrow$ faster tracking.
- Update must never increase P .

Deliverable: `gain_covariance_lab.py` + convergence plots and a Q comparison.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	The gain is a ratio of what to what?	State unc. / total unc.
Q2	Measurement noise H rises sharply. Which way does the gain move?	Down — trust the model
Q3	What two conditions make the Kalman filter optimal?	Linear + Gaussian

All three come from the gain formula, the trust dial, and the optimality conditions.

Three Misconceptions to Avoid

Where gain intuition slips

Common myths

- "You tune the gain by hand."
- "A higher gain is always better — it reacts faster."
- "The Kalman filter is always optimal."

The reality

- The optimal gain is computed from the covariances — no tuning.
- High gain reacts fast but inherits noise; there is a trade-off.
- Optimality needs linearity, Gaussian noise, and known parameters.

Three Things to Carry Forward

Part 2 in one screen

1

The gain balances model against data

It is a ratio of uncertainties: large when the state is uncertain or the data is sharp, small otherwise. It is computed, not tuned — the filter hands it to you.

2

The covariance drives everything

P grows in predict, shrinks in update, and feeds the gain each step. It gives calibrated error bars and, for time-invariant systems, converges to a steady state.

3

Five equations, optimal under clear conditions

Predict two, gain one, update two — the whole filter. It is the minimum-variance estimator when the system is linear and the noise is Gaussian.

Next — Part 3: Kalman in practice — denoising, a dynamic beta, and pairs trading Stage III.